BrainKAN Research Evidence Brief: Methodological Evaluation of Learned Spline Nonlinearities

Author: Xin Qi | **Production Commit SHA:** `4f6c2e8caba09fc7d0ddcda503da87f380424b3f`

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1. Research Motivation & Core Methodological Problem

In graph neural networks for functional neuroimaging, Kolmogorov–Arnold Networks (KANs) replace linear edge weights with learnable univariate B-splines $\Phi_{ij}(x) = \sum_p c_{ij,p} B_p(x)$, allowing direct visual inspection of edge-level transformations. In literature, researchers often interpret learned spline curvatures as evidence of in-vivo biological computation or task-specific cognitive mechanisms.

The Foundational Methodological Problem: *Do learned spline nonlinearities reflect task-specific information, or are they mathematical artifacts of operational metric definitions, evaluation domains, and non-convex optimization randomness?* We identify two primary unmodeled confounding factors:

1. **Reference-Domain Shifts:** Measuring curvature across shifted input supports creates apparent nonlinearity jumps purely from grid boundary extrapolation rather than model differences.
2. **Optimization Stochasticity:** Single initialization seeds can easily produce statistically significant conditional curvature differences that vanish or reverse across the broader optimization landscape.

2. Model Architecture: BrainKAN Edge-Spline Parameterization

The benchmark evaluates a specialized 2-layer Graph Kolmogorov–Arnold Network tailored for functional connectomes:

- **Layer 1 (Learnable Edge B-Splines):** For each directed functional edge $e = (u, v) \in \mathcal{E}$, message passing computes: $\Phi_e(x_u) = w_{\text{base},e} \cdot \text{SiLU}(x_u) + \sum_{k=1}^{G+k_o} c_{e,k} B_{k,k_o}(x_u)$, where B_{k,k_o} denotes cubic B-spline basis functions ($k_o = 3$) parameterized over $G = 5$ grid intervals ($G + k_o = 8$ basis functions).
- **Exact Parameter Accounting:** For $|E| = 240$ edges, $C_{\text{in}} = 1$, and $C_{\text{out}} = 16$:
  - Spline coefficients: $240 \times 16 \times 1 \times 8 = \mathbf{30,720}$ parameters.
  - Base residual weights: $240 \times 16 \times 1 = \mathbf{3,840}$ parameters.
  - Root self-connections: $16 \times 1 = \mathbf{16}$ parameters.
  - **Layer 1 Total: 34,576** parameters. The complete 2-layer BrainKAN model contains $P = \mathbf{589,779}$ parameters.
- **Operational Metric (Deviation-From-Linearity, DFL):** For any learned function f evaluated on discrete grid $D = \{t_1, \ldots, t_K\}$, DFL measures normalized variance unexplained by ordinary least-squares affine regression:

$$\text{DFL}(f; D) = \frac{\inf_{a,b} \sum_{k=1}^{K} |f(t_k) - (at_k + b)|^2}{\sum_{k=1}^{K} |f(t_k) - \bar{f}|^2} = \frac{\text{SS}_{\text{res}}}{\text{SS}_{\text{tot}}} = 1 - R^2$$

3. Dataset, Pairing Manifest, and Cross-Validation Integrity

- **Primary Cohort (HCP-WM):** 100 healthy young adults from the Human Connectome Project 3T Working Memory task-fMRI. Contrast: 2-Back (high cognitive load) vs. 0-Back (low cognitive load), comprising 200 total graphs (100 strictly paired observations).
- **Brain Graph Topology:** 48 cortical Regions of Interest (ROIs), 240 directed functional edges ($|E| = 240$).
- **Cross-Validation Integrity:** 5-fold CV with strict subject-level pairing (`data/subject_pairing_manifest.csv`). 0-Back and 2-Back scans for the same subject never cross fold boundaries. Fold reference grids $\mathcal{R}_k$ are computed exclusively from training subjects (zero train-test leakage).
- **External Cohort (CHCP):** Chinese Human Connectome Project dataset utilized for cross-cohort transportability checks.

4. Core Experimental Design: Progressive Falsification Suite

The benchmark evaluates 5 pre-committed experiments crossing input shifts with design-consistent null models:

- **Exp 1 (High-Precision Permutation Baseline):** Evaluates single-seed (`seed=42`) excess curvature on whole-graph average DFL over $N = 500$ sign-flip label permutations on $\mathcal{S}_{\text{pooled}}$ ($\Delta = +0.00162, p = 0.0060$). This global-graph test is distinct from the edge-level FDR test in exploratory Q3B (where 0/240 edges reached significance).
- **Exp 2 (Synthetic Ground-Truth Calibration):** 3-condition controlled simulation ($N = 200$) on known function $f^*(x) = x + 0.5\sin(2x)$ assessing DFL sensitivity to uniform, domain-shifted, and density-weighted grids.
- **Exp 3 (Reference Domain Control):** 5-fold CV on fMRI data contrasting pooled ranges against empirical common support $\mathcal{S}_{\text{common}} = \text{supp}(X_0) \cap \text{supp}(X_1)$ to eliminate boundary extrapolation.
- **Exp 4 (Capacity Sensitivity Analysis):** Multi-seed capacity sensitivity scan across BrainKAN hidden dimensions ($d \in \{4, 8, 16\}$ across 5 random initializations), with model parameter counts reported separately.
- **Exp 5 (Seed-Marginalized Permutation Testing, SMPT):** Multi-seed evaluation across $S = 10$ seeds $\times$ $B = 15$ permutations under **dual estimands**: (1) Pooled Support $\mathcal{S}_{\text{pooled}} = Q_{1\%-99\%}([X_0, X_1])$; (2) Strict Common Support $\mathcal{S}_{\text{common}} = \text{supp}(X_0) \cap \text{supp}(X_1)$.

5. Primary Empirical Findings Across Experimental Suite

Experiment	Estimand & Target	Sample Budget	Effect Size / Statistic	p-value & Uncertainty
Exp 1: Permutation	Single-seed ($s = 42$) on $\mathcal{S}_{\text{pooled}}$ (whole-graph)	$N = 500$ perms	$\Delta_{\text{pool}} = +0.00162$	$p = 0.0060$ ($k = 2/500$). Significant conditionally at graph level.
Exp 2: Synthetic	Grid gauge sensitivity on $f^*(x)$	$N = 200$ sims	Shifted $\Delta = +0.0816$	SD $= 0.0084$ ($> 40\times$ inflation purely from grid shift).
Exp 3: Common Support	Cross-cohort transfer (HCP→CHCP)	5-fold CV	$\Delta_{\text{common}} = -0.0003$	5-fold mean. Excess curvature collapses to zero.
Exp 4: Capacity Scan	BrainKAN hidden dims ($d = 4, 8, 16$)	5 seeds × 3 dims	Baseline $T \approx 0.176 - 0.183$	High test accuracy ($90\% - 93\%$). Baseline stable across widths.
Exp 5: Estimand 1	Multi-seed on Pooled Grid $\mathcal{S}_{\text{pooled}}$	$S = 10$, $B = 15$	$\bar{\Delta}_{\text{pool}} = \mathbf{-0.00058}$	95% t-CI: $[-0.00130, +0.00013]$, $p = 0.9513$ (1/10 pos).
Exp 5: Estimand 2	Multi-seed on Common Support $\mathcal{S}_{\text{common}}$	$S = 10$, $B = 15$	$\bar{\Delta}_{\text{common}} = \mathbf{+0.00090}$	95% t-CI: $[+0.00051, +0.00129]$, $p = 0.0003$ (9/10 pos).

6. Inferential Limitations & Methodological Boundary Conditions

To preserve rigorous scientific integrity, our findings are bounded by explicit inferential commitments:

1. **Optimization Sensitivity vs. Population Generalization:** Exp 5 evaluates $S = 10$ optimization initializations on a fixed cohort. It demonstrates **initialization consistency**, not population-level generalizability across unseen patient cohorts.
2. **Operational Residual Ratio vs. Differential Geometry:** DFL $= 1 - R^2$ is an operational metric of unexplained variance against affine regression over a discrete grid, **not** an intrinsic coordinate-invariant Riemannian curvature tensor.
3. **Finite Permutation Precision ($B = 15$):** The 10 seeds share the same 15 permutation flip vectors ($\text{SE}_{\text{MC}} \approx 0.00015$). Exp 1 provides a high-precision single-seed anchor ($B = 500, p = 0.0060$).
4. **Representation Inspection vs. Held-Out Generalization:** DFL inspects internal parameter geometry, not held-out classification accuracy. 5-fold CV functions as a subset sensitivity check.
5. **Empirical Fallback Numerical Guard Audit:** The common support guard (`if c_min >= c_max:`) was audited across all 5 folds × 48 ROIs (240 instances): 0/240 **triggered (**0.00%**)**. All reported values reflect genuine non-empty overlaps.

7. Programmatic Provenance & Codebase Path Index

Component	Repository Script / File Path	Role & Computational Output
Architecture	`models/kan.py`	Layer 1 B-spline EdgeConv (34,576 params) & BrainKAN (589,779 params).
Baselines	`models/baseline.py`	Canonical GCNBaseline (338 params) and GATBaseline (234 params).
Param Verification	`compute_baseline_params.py`, `tests/test_model_parameters.py`	Analytical/runtime parameter counting script and automated test assertions.
Pairing Manifest	`data/subject_pairing_manifest.csv`	100 paired HCP subjects with verified 5-fold CV assignments.
Exp 1 (Permutation)	`experiments/exp1_high_precision_permutation.py`	$N = 500$ permutation test → `results/exp1_high_precision/`
Exp 2 (Synthetic)	`experiments/exp2_synthetic_calibration.py`	Ground-truth gauge calibration → `results/exp2_synthetic/`
Exp 3 (Common Support)	`experiments/exp3_reference_domain_control.py`	Domain control & CHCP transfer → `results/exp3_ref_domain/`
Exp 4 (Capacity Scan)	`experiments/exp4_multi_seed_robustness.py`	Multi-seed hidden dim scan ($d \in \{4, 8, 16\}$) across 5 seeds.
Exp 5 (SMPT Multi-Seed)	`experiments/exp5_runner.py`, `run_exp5_parallel.py`	10-seed dual-estimand runner → `results/exp5_seed_stability/`
Support Overlap Audit	`experiments/compute_support_overlap_empirical.py`	Empirical overlap for 48 ROIs → `results/primary_analysis/`
Archived Pilot Runs	`results/archived_runs/pilot_unpermuted_diagnostic/`	Archived pre-test dry runs ($N_{\text{perms}} = 0$, effect = NaN).
Master Manifest	`results/manifest.json`	Git commit `4f6c2e8caba09fc7d0ddcda503da87f380424b3f` & canonical index.
Closed Audit Report	`docs/audit_report.md`	Internal audit closure with historical tracking and resolution commits.